

TED (15) 6025

(REVISION: 2015)

**SIXTH SEMESTER DIPLOMA MODEL EXAMINATION IN
ENGINEERING / TECHNOLOGY**

INDUSTRIAL AUTOMATION AND MECHATRONICS

(Time : 3 Hours)

(Maximum Marks: 100)

Part A

(Maximum Marks: 10)

I Answer all questions in one or two sentences. Each question carries equal marks.

1. Define automation
2. Define the term, time constant of a sensor.
3. List out any 4 displacement sensors
4. Draw the symbol of a pressure sequence valve
5. Differentiate between micro controller and micro processor.

5 x 2 = 10

Part B

(Maximum Marks: 30)

II Answer any five questions. Each question carries 6 marks.

1. Explain the elements of a measurement system
2. Explain the term non- linearity error in sensors
3. Explain how an LVDT be used for measuring the pressure
4. Illustrate the working of a triac, with its voltage current characteristics.
5. Explain the working of a variable reluctance stepper motor.
6. Draw a ladder diagram to continuously on-off an output at some preset time
7. Explain the jump control in PLC programming with an example

5 x 6 = 30

Part C

(Answer one full question from each unit. Each question carries 15 marks.)

Unit 1

- III** (a) Explain the elements of a control system with a suitable example **8**
- (b) Explain the advantages and disadvantages of automation **7**

OR

- IV** (a) Differentiate between fixed automation and flexible automation **8**
- (b) Explain a mechatronics system with a block diagram **7**

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Unit 2

- V** (a) Explain how a potentiometer sensor is used for measuring displacements. 8
(b) Illustrate the working of Hall effect sensor as a fuel level indicator in vehicles. 7

OR

- VI** (a) Explain the different type of fluid flow sensors. 8
(b) Explain the different type of light sensors 7

Unit 3

- VII** (a) Explain the use of a pressure relief valve in a hydraulic system with a neat sketch. List out its functions 8
(b) Explain the working of a bi-polar transistor as a solid state switch. Also draw its voltage current characteristics. 7

OR

- VIII** (a) Explain with a block diagram, the working of a pneumatic actuation system. Also draw the circuit with symbols. 8
(b) Illustrate the working of a solenoid operated direction control valve 7

Unit 4

- IX** (a) Explain the different data handling instructions in PLC ladder diagram 8
(b) Compare the mechanical and mechatronics design solutions for a bath room scale. 7

OR

- X** (a) Explain the fault detection techniques in microprocessor 8
(b) Write a ladder program to operate 3 output lamps from the following data. 4 inputs A, B, C and D are connected to the PLC. The output lamp 1 should switch on if there are at least two inputs. The lamp 2 should switch on only if there are three inputs. If all the inputs are on, lamp 3 should be switched on. 7